25900746 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION NOVEMBER 2025

FIRST SEMESTER

Multi-Disciplinary Courses (MDC) - MG1MDCBIF100 - BIOINFORMATICS FOR BEGINNERS

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 35**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **5** out of **7** Questions

Each Question carries **2** marks

1. Interpret hierachial organization of life. [U] / [CO1, CO2]
2. Relate the purpose of global alignment in bioinformatics. [A] / [CO6]
3. Which organization manages the Protein Data Bank (PDB)? [K] / [CO5, CO6]
4. Name the primary function of Clustal Omega. [K] / [CO6]
5. Draw a simple representation of a DNA sequence with its complementary strand. [K] / [CO6]
6. Cherish the important process involved in central Dogma of biology [Ap] / [CO1, CO2]
7. How has the Human Genome Project impacted medicine? [K] / [CO4]

[5x2 = 10]

Part B

Short Essay Type Questions

Answer any **3** out of **5** Questions

Each Question carries **5** marks

8.	Correlate the role of bioinformatics in drug discovery and development.	[An] / [CO6]
9.	Uphold the aim and scope of bioinformatics.	[Ap] / [CO3]
10.	Illustrate the process of retrieving data from UniProt?	[U] / [CO5, CO6]
11.	Define pairwise sequence alignment and explain its significance in bioinformatics.	[U] / [CO6]
12.	Analyze the significance of structural databases in understanding protein function and interactions.	[U] / [CO5, CO6]

[3x5 = 15]

Part C

Long Essay Type Questions

Answer any **1** out of **3** Questions

Each Question carries 1**0** marks

13.	Define molecular visualization and explain its significance in bioinformatics.	[U] / [CO6]
14.	Explore biomolecules and its types.	[I] / [CO1, CO2]
15.	Embrace the basic difference in computational biology and bioinformatics.	[I] / [CO3]

[1x10 = 10]